ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Differentiation

Differentiate to read gradient, tangent, normal, and stationary behaviour.

Differentiation · P1 · 12

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA11 website answer bank. The topic reduces to these exam moves.

01 POWER: Differentiate Powers
TRIGGER *find* $\frac{dy}{dx}$

02 TANG: Tangent Line
TRIGGER *tangent at a point*

03 NORM: Normal Line
TRIGGER *normal at a point*

04 STAT: Stationary Points
TRIGGER *stationary point, increasing/decreasing*

BANK EVIDENCE Local pages: 27 problem, 54 answer, 17 notes. Website primary entries: 24.

01 POWER: Differentiate Powers

TRIGGER *find* $\frac{dy}{dx}$

BECOMES Bring the power down and reduce it by 1.

FIRST LINE TO WRITE

$$\frac{d}{dx}(ax^n) = anx^{n-1}$$

SIMPLEST STRATEGY

1 **P**ut terms into power form.

2 **O**perate: multiply by the power.

3 **W**rite the new coefficient.

4 **E**xponent decreases by 1.

5 **R**ewrite neatly.

WORKED MODEL

$$\frac{d}{dx}(3x^4 - 2x^{1/2}) = 12x^3 - x^{-1/2}.$$

HARD VARIANTS

1 Rewrite roots and denominators as powers before differentiating.

2 If the expression is expanded brackets over roots, simplify first.

3 Constants differentiate to zero, but parameters multiply like constants.

BOTTOM LINE

Rewrite roots and fractions before differentiating.

Practice 01

SAME IDEA Use POWER: Differentiate Powers.

QUESTION

> Differentiate $5x^3 - 4\sqrt{x}$.

YOUR SOLUTION

02 TANG: Tangent Line

TRIGGER *tangent at a point*

BECOMES Differentiate, substitute x, then use line equation.

FIRST LINE TO WRITE

$$m = f'(a)$$

SIMPLEST STRATEGY

1. **T**ake derivative.
2. **A**t the point, find gradient.
3. **N**ame the point.
4. **G**enerate $y - y_1 = m(x - x_1)$.

WORKED MODEL

$$f'(2) = 5,\ P(2,7) \Rightarrow y - 7 = 5(x - 2).$$

HARD VARIANTS

1. If only x is given, find y from the original curve first.
2. For parallel tangents, set $f'(x)$ equal to the required gradient.
3. For parameter tangents, use both point-on-curve and gradient conditions.

BOTTOM LINE

A tangent is just a line with gradient $f'(a)$.

Practice 02

SAME IDEA Use TANG: Tangent Line.

QUESTION

> Find the tangent to $y = x^2 + 1$ at $x = 3$.

YOUR SOLUTION

03 NORM: Normal Line

TRIGGER *normal at a point*

BECOMES Use the negative reciprocal of tangent gradient.

FIRST LINE TO WRITE

$$m_n = -\frac{1}{m_t}$$

SIMPLEST STRATEGY

1. **N**eed tangent gradient first.
2. **O**btain negative reciprocal.
3. **R**ecord point.
4. **M**ake the line equation.

WORKED MODEL

If $m_t = 4$, then $m_n = -\frac{1}{4}$.

HARD VARIANTS

1. Find the tangent gradient first, then use the negative reciprocal.
2. If tangent gradient is zero, the normal is vertical: $x = a$.
3. If tangent is vertical, the normal is horizontal.

BOTTOM LINE

Normal gradient is negative reciprocal.

Practice 03

SAME IDEA Use NORM: Normal Line.

QUESTION

Find the normal gradient when $f'(a) = -2$.

YOUR SOLUTION

04 STAT: Stationary Points

TRIGGER *stationary point, increasing/decreasing*

BECOMES Set derivative equal to zero.

FIRST LINE TO WRITE

$$f'(x) = 0$$

SIMPLEST STRATEGY

1. **S**et derivative to zero.
2. **T**ake roots.
3. **A**ssess sign or second derivative.
4. **T**ell max/min/increasing/decreasing.

WORKED MODEL

$$f'(x) = 2x - 6 = 0 \Rightarrow x = 3.$$

HARD VARIANTS

1. Stationary points come from $f'(x) = 0$, then y from the original curve.
2. Use sign change or second derivative to classify max/min.
3. For increasing/decreasing intervals, test the derivative sign between critical values.

BOTTOM LINE

Stationary means gradient zero.

Practice 04

SAME IDEA Use STAT: Stationary Points.

QUESTION

> Find stationary points of $y = x^2 - 8x + 1$.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
find $\frac{dy}{dx}$	$\frac{d}{dx}(ax^n) = anx^{n-1}$
tangent at a point	$m = f'(a)$
normal at a point	$m_n = -\frac{1}{m_t}$
stationary point, increasing/decreasing	$f'(x) = 0$